Fadhil Muhammad

Software Engineer & IT Consultant · Palm oil and plantation systems

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Portfolio and case studies: pamulab.vercel.app/work

SUMMARY

Software engineer and IT consultant with 3+ years building mobile, web and backend systems. I lead GEMA Plantation and GEMA Factory at Green Harvest Solutions: offline-first field apps, web CMS, .NET Core APIs and ERP integration for palm oil estates and mills. I work on site with clients, from field studies and requirement gathering to delivery, user training and support, and I lead a small development team. Former Flutter facilitator for BEKUP × Dicoding.

SKILLS

Mobile	Kotlin, Hilt, Room, CameraX, NFC, Flutter, Provider, Dio, Freezed
Web	Next.js, React, TypeScript, Tailwind CSS, TanStack Query, Motion, Leaflet
Backend & infra	.NET Core, SQL Server, ASP.NET Web Forms, Crystal Reports, Node.js, Prisma, Docker, IIS, Plesk, NGINX, GitHub Actions
AI & monitoring	TensorFlow Lite, YOLOv8, LLM APIs, Prometheus, Grafana, Sentry, Crashlytics, Postman, Swagger
Analysis & design	User flows, ERD, Wireframes, Figma, Canva, Photoshop, Notion
Ways of working	Field studies, requirement gathering, estimation, Agile delivery, code review, user training, client presentations

EXPERIENCE

Software Engineer · Green Harvest Solutions · PT Semesta Agri Solusi — Dec 2025 – Present

Medan · Full-time · On site and with clients

Mobile and web software for palm oil operations: the GEMA product line and the Green Harvest Plantation ERP, deployed per client with isolated servers and databases.

- Lead GEMA Plantation and GEMA Factory with a team of usually two (up to four) developers, from field studies and requirement gathering to estimates, presentations and delivery.
- Built GEMA Factory, a Flutter mill maintenance app: preventive maintenance, work orders, approvals, workshop activities, reporting and role management.
- Develop the GEMA Plantation Android app (Kotlin/XML, offline-first, on-device ripeness detection, NFC) and its CMS in Next.js, TypeScript, Tailwind CSS and TanStack Query.
- Integrated an AI chatbot that answers questions about production, maintenance and reports through API-driven access to operational data.
- Build .NET Core and SQL Server APIs (documented with Swagger), and support client ERP systems on ASP.NET Web Forms, Crystal Reports and IIS.
- Deploy and monitor client servers (IIS, Plesk, Prometheus, Grafana), handle production troubleshooting, and run the Git and GitHub workflow.
- Train users and support clients online and on site; rebuilt the company website into a product site and design product decks and brochures.

Software Engineer · Samta Cloud — Nov 2025 – May 2026

Medan · Part-time · Project-based

- Developed a Flutter app for hospitality and hotel maintenance operations, digitising day-to-day maintenance workflows.
- Owned UI implementation, REST API integration and state management, using clean architecture to keep the app maintainable.
- Worked with the development team to keep the app stable and aligned with business requirements.

Product Engineer · Freelance — Nov 2023 – Apr 2026

Remote · Self-employed

- Worked directly with clients to find the core problem and turn business needs into working software.
- Analysed and built a purchasing, sales and inventory (P3) system for a POS and stock business: process diagrams, wireframes, data model, then full-stack build.
- Delivered a mobile attendance app for a private company with location-based validation to keep presence records accurate.
- Designed a livestock and distributor management system and a multi-role preloved marketplace in Figma, from system architecture and user flows to hi-fi prototypes.

Flutter Facilitator · BEKUP × Dicoding Indonesia — Jul 2025 – Nov 2025

Remote · Part-time

- Facilitated weekly online sessions and tracked progress for participants from very different backgrounds.
- Reviewed and validated participant journals with 100% compliance to programme standards.
- Supported 15+ instructor-led training sessions end to end.
- Introduced peer programming and collaborative learning to strengthen retention and teamwork.

Flutter Developer · DMB · PT Damai Nusa Abadi (DNA Group) — Oct 2024 – Sep 2025

Medan · Hybrid · Full-time

- Designed and built the self-ordering app for Android TV kiosks and tablets: UI/UX, API integration and state management.
- Built attendance with facial recognition and fake-GPS detection, notifications, and offline mode for POS.
- Delivered modules for hotel, asset, POS, inventory and multi-outlet management systems.
- Released and maintained the group's apps on Google Play; worked with cross-functional teams to meet stakeholder requirements.

Graphic Designer (Internship) · Sentosa Digital Printing — Aug 2019 – Nov 2019

Medan · Vocational school field practice

- Handled graphic design requests directly with clients, adapting to different client types and briefs.

LEADERSHIP & COMMUNITY

Master of Ceremonies, Universitas Mikroskil (Nov 2023): Hosted the seminar “Cerdas Emosi Raih Prestasi” (Emotional Intelligence Leads to Achievement) for an audience of university students.

President, BITSMIKRO, Informatics Student Association (Sep 2022 – Sep 2023): Led and managed the association and its divisions for a full term.

Mentor Assistant, Universitas Mikroskil (Jan 2023): Helped participants solve the problems they ran into during the hands-on sessions.

Peer Tutor, Study Group, Kampus Merdeka × Dicoding, Studi Independen (Aug 2022 – Dec 2022): Taught and assisted fellow participants with the problems they ran into during the Flutter and back-end learning path.

Media & Information Division Staff, BITSMIKRO, Informatics Student Association (Dec 2021 – Sep 2022): Communicated the association's news through social media posts, flyers, thumbnails and the organisation blog.

EDUCATION & TRAINING

Universitas Mikroskil, Bachelor of Informatics Engineering · GPA 3.91 / 4.00. Thesis project: Angkutin, a waste pickup app — 2020 – 2024

Bangkit Academy (Google, GoTo, Traveloka), Mobile Development cohort · Kotlin, Jetpack Compose, SOLID principles and UX design, plus a capstone with ML and cloud teams — Feb 2023 – Aug 2023

Kampus Merdeka × Dicoding, Studi Independen, Multi-Platform (Flutter) & Back-End (Node.js) · Best Graduate, and peer tutor for the study group — Aug 2022 – Dec 2022

SMK Negeri 10 Medan, Multimedia — 2017 – 2020

CERTIFICATIONS & LANGUAGES

Flutter Developer Expert, Dicoding Indonesia, 2022 · Android Developer (Jetpack Compose), Bangkit Academy, 2023 · Back-End Developer (Node.js), Dicoding Indonesia, 2022 · Belajar Penerapan Machine Learning untuk Flutter · Learn SOLID Programming Principles · Learn Git Basics with GitHub

Indonesian (native) · English (limited working proficiency)